



# Scaling Up Private LLM Agents to Synthesize Household Energy Consumption Data

Thi Ngan Dang (Phoebe), Jasmine Huynh, Mahathir Almashor



## Introduction

Acquiring large-scale, real-world household energy data is challenging due to privacy concerns.

### Research Question:

Can we generate a larger, more realistic energy dataset using a private LLM on HPC infrastructure?



## Methodology

Simulated Agents using household appliances



**Data Generation:** Use private LLMs to simulate household activities and generate realistic energy consumption patterns.

### Limitations of Previous Methods:

- Unable to run on HPC infrastructure.
- Could only generate data for four households.

### Scaling the Process:

- Leveraged **Pawsey's Setonix supercomputer** (High-performance computing - HPC) and **AMD GPU** to scale up to a larger scope.
- Chose private LLMs for efficiency, security, and sustainable computing.

**Validation:** Initial verification of daily generated data against real-world data per households.

## Implementation

**LLM Evaluation:** Researched and compared private LLM models (**Mistral-7B**, Minstral-8B, Llama3.2)

### HPC Integration:

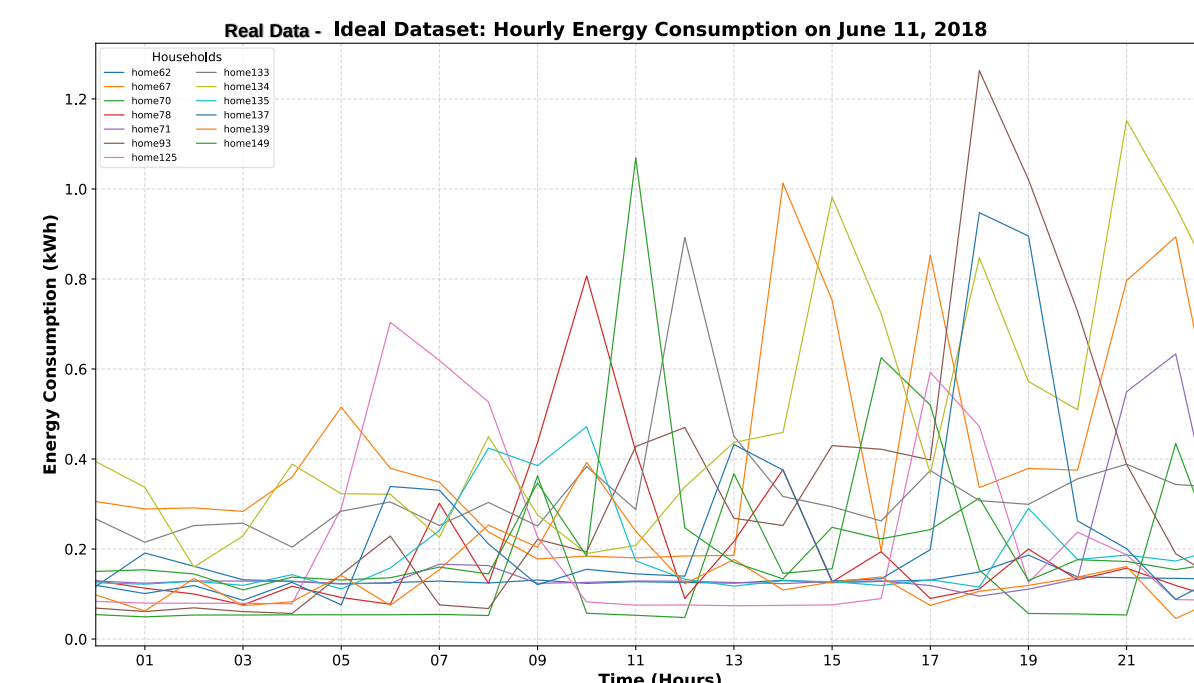
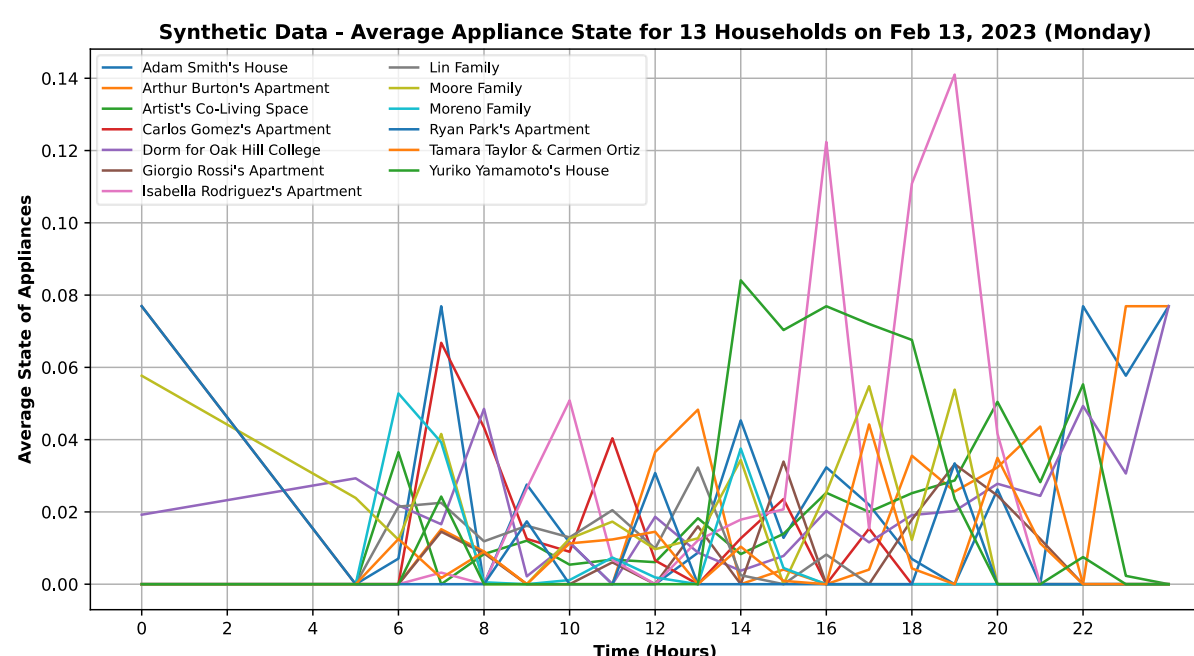
- Built proof-of-concept components to validate the feasibility for deploying LLMs and web servers on Setonix.
- Adapted existing codebase not designed for HPC infrastructure through systematic debugging and testing protocols
- Implemented secure port forwarding solution to enable web server functionality and resolved API connection issues between frontend and backend systems

### Performance Optimization:

- Optimized LLM prompt design for precision
- Benchmarked across various GPUs and time granularity to speed up data generation without compromising quality.

## Results

- **Successfully deployed** the data generation process on Setonix.
- Initial step produced synthetic energy consumption data for **13 households and 25 agents** over **24 hours** (30 second time granularity).
- The generated data shows **randomness** and **does not follow a uniform pattern**, that looks like real-world data
- **Explainability:** Compared to real-world data (privacy-protected), we can analyze the simulation results to investigate why certain households have higher energy usage.



## Conclusion

- Demonstrated the proof-of-concept and potential of combining private LLMs and supercomputing to generate scalable, realistic datasets.
- Conducting multiple data generation runs at varying time intervals (10s, 30s, 60s) and across different GPUs allowed us to evaluate their performance, refining our approach for future scalability

### Limitation:

- Increasing time granularity to 60s introduces time gaps in the synthetic data.
- Reducing the interval improves data consistency but increases generation time.
- Due to this project time constraints, a trade-off between data quality and efficiency was necessary.

### Future Work:

- Investigate methods to scale up to a larger size of households
- Hyper-localize datasets for regions like Hanoi or Jakarta to capture diverse energy behaviors.
- Explore with the latest private LLMs to increase accuracy such as DeepSeek-V2 Lite, Microsoft's Phi-3 ...
- Aim to improve data consistency

## References

[Link to References](#)